

3. Suppose A is an $m \times n$ matrix, all of whose rows are identical.
 Suppose B is an $n \times p$ matrix, all of whose columns are identical.
 What can be said about the entries in AB ?

Solution: Since each row of A is identical, let any arbitrary row i be $[a_1 \ a_2 \ a_3 \ \dots \ a_n]$.

Since each column of B is identical, let any arbitrary column j be $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$.

By Row-Column Rule for Computing AB ,

$$(AB)_{ij} = a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n$$

Any arbitrary element of AB will be the above value. This is true since our choice of i and j are arbitrary.

$\therefore$ Each entry in AB are equal $\square$

Exercises 2.1

In Exercises 1 and 2, compute each matrix sum or product if it is defined. If an expression is undefined, explain why. Let

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & -5 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & -5 & 1 \\ 1 & -4 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 5 \\ -1 & 4 \end{bmatrix}, \quad E = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$$

1. $-2A, B-2A, AC, CD$
 $-2A = -2 \begin{bmatrix} 2 & 0 & -1 \\ 4 & -5 & 2 \end{bmatrix} = \begin{bmatrix} -4 & 0 & 2 \\ -8 & 10 & -4 \end{bmatrix}$ (By defn. of scalar multiple)

$$B-2A = B + (-2A) = \begin{bmatrix} 7 & -5 & 1 \\ 1 & -4 & -3 \end{bmatrix} + \begin{bmatrix} -4 & 0 & 2 \\ -8 & 10 & -4 \end{bmatrix} = \begin{bmatrix} 3 & -5 & 3 \\ -7 & 6 & -7 \end{bmatrix}$$
 (By defn. of sum)

A has 2 rows, 3 columns. C has 2 rows, 2 columns.
 $\therefore$ The no. of columns of A does not match the no. of rows of C ,

$\therefore$ by defn, AC is not defined.

D has 2 rows and 2 columns. The no. of columns of C matches the no. of rows of D . $\therefore$ By defn of matrix multiplication CD is defined.

$$CD = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$